



**Dept. of Computer Science &
Engineering(AI&ML) and Computer Science &
Engineering(Cyber Security)**

**Programme : B E – Computer Science & Engineering(AI&ML) and Computer Science &
Engineering(Cyber Security)**

Assessment – 1

Q. NO	Questions	Blooms Levels	CO	Marks
1.	Solve the following i. Prove that $f(n) = 4n^2 + 64n + 288$ is $O(n^2)$. Define Big-Oh Notation and Compute the value of c and n_0 , which holds the Notation true. ii. Consider the following claims I. $(n+k)^m = \Theta(n^m)$, where k and m are constants II. $2^{n+1} = O(2^n)$ III. $2^{2n+1} = O(2^n)$ Which of these claims are correct? Justify your answer.	L4	1	2
2.	Design an algorithm for checking whether two given words are anagrams, i.e., whether one word can be obtained by permuting the letters of the other. Example: tea and eat are anagrams. Calculate the Total Running time of the algorithm design based on counting the primitive operations.	L4	1	2
3.	Apply Quick Sort to sort the list 34,23,22,45,25,38,28,21,29,35,33 in ascending order. Fix the pivot position as last element of the array.	L3	2	2
4.	Design an algorithm for solving the problem given below: Consider this problem: You have a row of binary digits arranged randomly. Arrange them in such an order that all 0's precede all 1's or vice-versa. The only constraint in arranging them is that you are allowed to interchange the positions of binary digits if they are not similar. Illustrate the algorithm process with the example given below 11010100 $\rightarrow$ 00001111	L4	1	2



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5.	Apply Masters Theorem and compute the time complexity for the recurrence relations i. $T(n)=9T(n/3)+n^2$ ii. $T(n)=2T(n/2)+\frac{n}{\log n}$	L4	1, 2	2
6.	Apply Merge Sort to sort the list E, X, A, M, P, L, E in alphabetical order. Write the recurrence relation for Merge sort.	L3	3	2
7.	Analyse and find the time complexity of the below function code snippet. i. <pre>void fun1(){ for(int i=1;i<=n;i++){ cout<<i<<endl; } for(int j=1;j<=m;j++){ cout<<j<<endl; } }</pre> ii. <pre>void fun2(int n){ for(int i=0;i<n;i++){ cout<<"At MSRIT"; } }</pre> iii. <pre>void fun3(){ for (int i=0;i<n;i++){ int a=5; a++; } for (int i=0;i<n;i++){ for(int j=0;j<n;j++){ int b=5; b++; } } }</pre>	L4	1	3